

Discovery Executive Summary

Project: PingCRM-Discovery · Generated: 30/09/2026, 11:39:55

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating
1	Code Quality & Complexity Analysis	—

1. Code Quality & Complexity Analysis

EXECUTIVE SUMMARY

The PingCRM codebase suffers from extreme code duplication that dwarfs all other quality concerns. Approximately 70.5% of the 187,334 source lines are near-identical copies across 80 IVR controllers (759 LOC each, differing only by module name), 12 "GodService" classes, 8 legacyFormatters files, 133 LegacyPass2 React page components, and 147 legacy class widgets. The frontend layer (1,050 files / 107,920 LOC) is disproportionately affected, with duplicated utility files exceeding 1,000 LOC each and 133 single-function components at 387 LOC apiece. Backend duplication centres on a mechanically stamped controller-per-action pattern where every IVR module has 7 identical controllers. Cyclomatic complexity per method is moderate (max ~18), but the sheer volume of copied code makes safe modification nearly impossible. Git churn is low (3 commits in 6 months), indicating the IVR layer was bulk-generated and has not yet been iterated on — the duplication debt will compound the moment feature work begins. Both layers were analysed: 180 PHP files (backend) and 1,050 TS/TSX/JS/JSX files (frontend).

2.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured	Rating
H1	High Cyclomatic Complexity	Max complexity per method	<10	10–20	>20	~18 (Hub/Index.tsx component)	Moderate
H2	Large Classes	Largest class/file LOC	<300	300–1000	>1000	1,101 LOC (legacyFormatters*.ts — 8 files)	High Risk
H3	Large Functions	Largest function LOC	<50	50–200	>200	387 LOC (LegacyPass2 components — 133 files)	High Risk
H4	Business Logic Duplication	Duplicated business logic %	<5%	5–10%	>10%	~34% (GodServices + IVR controllers)	High Risk
H5	Duplicate Code (general)	Overall duplicate code %	<5%	5–10%	>10%	~70.5% (132,077 of 187,334 LOC)	High Risk
H6	High Churn Areas	Monthly changes (top files)	<5	5–10	>10	<2/month (top file: composer.lock — 27 all-time)	Good
H7	Defect-Prone Files	Fix commits (hottest file)	1–3	4–5	>5	2 (HandleInertiaRequests.php, Users/Edit.vue)	Good
H8	Ownership Issues	Top-author ownership %	>80%	60–80%	<60%	100% single-author (IVR layer); 52% top-author overall	Good
H9	Unsafe Dynamic Variables (additional)	Files using <code>extract()</code> on user input	0	1–5	>5	92 files (80 controllers + 12 GodServices)	High Risk

Hotspot Score breakdown

Component	Weight	Sub-score (0–100)	Weighted
Cyclomatic Complexity	25%	50	12.5
Code Churn	25%	10	2.5
Defect Density	20%	15	3.0
Class/Function Size	15%	75	11.25

Business Logic Duplication	10%	95	9.5
Developer Ownership Risk	5%	10	0.5
Hotspot Score	100%		39 / 100

2.5 Actions Required

Hotspot	Action	Rating	Priority
H2 — Large Classes	Replace 8 legacyFormatters files with single parameterised function; consolidate 80 IVR controllers into 1 generic controller; merge 12 GodServices into 1 service	High Risk	Critical
H3 — Large Functions	Replace 133 LegacyPass2 components with single data-driven component; split Hub/Index.tsx into sub-components	High Risk	Critical
H4 — Business Logic Duplication	Create module registry config; implement single IvModuleService with Command pattern for workflows	High Risk	Critical
H5 — Duplicate Code (general)	Consolidate all legacy frontend copies (formatters, Pass2 pages, class widgets, hooks) into parameterised originals	High Risk	Critical
H9 — Unsafe Dynamic Variables	Replace <code>extract(\$payload)</code> with explicit assignment; add Form Request validation; replace raw SQL with query builder	High Risk	High
H1 — High Cyclomatic Complexity	Decompose Hub/Index.tsx into sub-components; extract LoadsvrModuleData query patterns into service	Moderate	Medium

2.6 Expected Outcomes

- **~70% reduction in codebase size** — eliminating duplicated code reduces ~187,000 LOC to ~55,000 LOC, dramatically lowering maintenance burden and build times.
- **Safer modifications** — a single IvModuleService and IvModuleCrudController means business-rule changes are made once, not 92 times.
- **Improved static analysis** — removing `extract()` and raw SQL enables PHPStan/Psalm and ESLint to catch real bugs; currently these tools cannot trace variable origins.
- **Faster onboarding** — new developers can understand 1 parameterised pattern instead of navigating 80 identical controller files.
- **Smaller frontend bundle** — replacing 133 LegacyPass2 components and 147 class widgets with data-driven alternatives reduces JS bundle size by an estimated 60+ KB
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